RODRIGO GOMES DE ARAÚJO

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Summary

MSc Computer Science student (graduating **July 2027**) focused on **systems programming, security and software correctness**. Co-founder and CEO of **Tinker**, an early-stage startup, leading product direction and hands-on engineering as its primary developer. Built a **proof-of-work blockchain** in Rust; current research extends **session types** to enforce correct use of cryptographic resources at compile time. Currently focused on **growing Tinker** and seeking **early-stage funding** to expand development.

Education

Master's — **Sep 2025 – Present (Jul 2027)**
Computer Science — *University of Porto (FCUP)*

Bachelor's — **Sep 2022 – Jul 2025**
Computer Science and Engineering — *University of Porto (FEUP)*

Experience

Tinker — **Jul 2026 – Present**
Co-founder, CEO & Lead Engineer

- Originated the product concept and co-founded Tinker with two partners. Lead company strategy, product direction and engineering, owning the architecture and serving as the primary developer.
- Designed and built a **lightweight Rust backend** integrating version control, project management and CI into a unified development workflow.
- Implemented the platform's **core storage components**: a Git underlay, content-addressed persistence, a permission store and replay mechanisms.
- Designed the **authorization and trust model**, with signed approvals bound to exact actions, tiered permissions and a tamper-evident history linking changes to their verification and approval.
- Built a **modular verification engine** and a **fully asynchronous, automated pipeline** coordinating implementation, verification, CI and review, including support for agentic implementation and concurrent verification of immutable versions.
- Developed a **VS Code extension** exposing the full development workflow within the editor, from tickets and specifications through implementation, verification and review.

Research

CaST (Cryptographically-aware Session Types) — **Jul 2026 – Present (Jul 2027)**
Master's Thesis — *FreeST, Haskell, Type Systems, Applied Cryptography*

- Extending the expressiveness of session types to capture **cryptographic resource usage**, so that its misuse in protocols is detected and rejected at compile-time.
- Formalizing the extension and providing a **soundness proof**, integrating it into the FreeST compiler.

Coursework

PoW Blockchain for Decentralized Auctions
Data Systems Security · 19/20 — *Rust, Tokio, LibP2P, Docker, Blake2b, Ed25519*

- Extensive **research on the state of the art** of Bitcoin, Ethereum and Solana and investigation regarding the security and resilience to failure of these systems.
- Developed a DHT overlay based on Kademlia for **decentralized network construction**.
- PoW Blockchain with **Blake2b hashing**, **Ed25519 signatures** and **Merkle Trees** for transaction validation.
- **Client for auctions** using the blockchain layer as a public ledger.
- Docker **adversarial network simulation** with clients and bootstrap, well behaved and malicious nodes.

Awards

2nd Place in IEEE RetroJam 2025 | IEEE UP Student Branch — **Oct 2025**

- Game from scratch in 48 hours (Rust and Raylib), learning how to **reduce a problem to its core requirements**.

Academic Merit Awards — **2020, 2021, 2022**

1st Place in the National Robotics Championship | Instituto Politécnico de Bragança — **May 2016**

Technical Skills

Languages: Rust, C, Python, SQL, Haskell, C++, Java
Core Areas: Systems Programming, Distributed Systems & P2P Networking, Type Systems & Formal Methods, Functional Programming, Applied Cryptography
Tech Stack: Tokio, Dioxus, libp2p, Tonic (gRPC), Git, Linux, Docker, Terraform
Data: PostgreSQL, PostGIS, pandas, NumPy
Spoken Languages: Portuguese (Native), English (Fluent), Spanish (Intermediate)